

Comprehensive Survey of Retrieval-Augmented, Knowledge-Graph, and Multimodal Large Language Models for Inclusive Healthcare Guidance: Architectures, Benchmarks, and Clinical Deployment

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Abstract—This paper surveys and compares Retrieval-Augmented Generation (RAG), Knowledge-Graph (KG) enhanced reasoning, and Multimodal Large Language Models (LLMs) within medical contexts. Beyond model accuracy, it emphasizes inclusivity, multilingual access, and doctor-in-the-loop feedback. Synthesizing 41 key publications (2018–2025), it covers architectures, datasets, reasoning mechanisms, safety metrics, and future challenges. The goal is to guide design of transparent, clinically validated, and equitable AI systems deployable in low-resource healthcare settings.

Index Terms—Retrieval-Augmented Generation, Knowledge Graph, Multimodal LLMs, Healthcare AI, Hallucination Mitigation, Fairness, Evaluation, Doctor-in-the-Loop

I. INTRODUCTION

The rapid digitization of healthcare—from electronic health records (EHRs) to imaging and patient-generated text—has created immense opportunities for AI-driven decision support. Transformer-based LLMs (GPT, BioBERT, ChatDoctor) show remarkable reasoning capacity yet exhibit three major weaknesses:

- 1) factual hallucination and outdated knowledge,
- 2) weak grounding on structured medical entities, and
- 3) linguistic bias against low-resource languages.

Recent research therefore pivots toward augmented architectures:

- **Retrieval-Augmented Generation (RAG)** adds document evidence from trusted corpora [3], [29].
- **Knowledge Graph Augmentation** introduces structured reasoning paths via medical ontologies (UMLS, Drug-Bank) [15], [31].
- **Multimodal Fusion** integrates images, waveforms, and clinical tables for holistic analysis [2], [8].

The confluence of these trends points toward clinically aware assistants capable of transparent and multilingual reasoning.

II. SURVEY SCOPE AND METHODOLOGY

This review covers 41 primary sources spanning IEEE Xplore, PubMed, ACL Anthology, and Nature Digital Medicine (2018–2025). Inclusion criteria:

- explicit mention of medical or clinical LLMs;
- quantitative evaluation on public or proprietary datasets;
- discussion of retrieval, KG, or multimodal fusion.

We extracted four feature sets per paper:

- 1) model architecture (retriever, encoder, generator);
- 2) dataset origin and modality;
- 3) evaluation metrics (accuracy, F1, BLEU, hallucination);
- 4) alignment or safety methods (e.g., RLHF, human review).

Qualitative sources were retained when introducing novel interpretability or fairness frameworks.

A. Survey Timeline

- **2018–2020:** Domain-specific pretraining (BioBERT, ClinicalBERT).
- **2021–2022:** Instruction tuning (ChatDoctor, BioInstruct).
- **2023–2024:** Retrieval and KG fusion (FreshLLM, Med-KGB).
- **2024–2025:** Vision-language and multilingual LLMs (Health-LLM, LLaVA-Med, Indic-Transformers).

III. BACKGROUND AND EVOLUTION OF MEDICAL LLMs

A. Stage 1 — Domain Pretraining

BioBERT and ClinicalBERT adapted BERT on PubMed and MIMIC-III to capture biomedical vocabulary [4], [6]. They improved NER and QA by 10–15 points over generic BERT but lacked dialogue coherence.

B. Stage 2 — Instruction and Dialogue Tuning

ChatDoctor [3] and BioInstruct [30] introduced medical instruction datasets (~200k prompt–response pairs). Fine-tuning

yielded context-aware dialogue and improved BLEU scores by 0.08–0.1.

C. Stage 3 — Retrieval and Structured Reasoning

FreshLLM and MedKGB extended generation pipelines with retrieval and graph reasoning layers, reducing hallucination by up to 60 %. This phase also saw the rise of *Self-RAG* and *KG-fusion* approaches for evidence-aware output.

D. Stage 4 — Multimodal and Cross-Lingual Expansion

Models like Med-Flamingo and Health-LLM aligned image embeddings with textual features for report generation. Indic Transformers and Bhasha-GPT extended coverage to Indian languages, reducing linguistic inequality [25], [26].

IV. TAXONOMY OF TECHNIQUES

Table I summarizes five families of medical LLM architectures and their trade-offs.

Each family represents a distinct axis of innovation: factuality (RAG), explainability (KG), perceptual breadth (multimodality), and social equity (multilinguality).

V. DATASETS AND EVALUATION PROTOCOLS

Reliable comparison demands shared benchmarks (Table II). Medical datasets span text, tabular, and visual modalities.

Evaluation dimensions include factual accuracy, hallucination rate, clinical relevance, and human expert review [18], [24], [34]. For multimodal tasks, additional grounding and localization metrics are used to assess alignment between image regions and generated text.

A. Emerging Evaluation Trends

- **Factuality Benchmarks:** MedRAG-Eval and FaithMed quantify evidence consistency.
- **Safety Audits:** harmfulness rate and clinician re-evaluation percentage.
- **Reasoning Metrics:** Chain-of-Thought (COT) coherence and step validity [27].

VI. QUANTITATIVE COMPARISON SETUP

Before aggregating results (§VIII), we define evaluation axes:

- 1) Accuracy and F1 for clinical QA;
- 2) BLEU / ROUGE-L for summarization;
- 3) Hallucination percentage for factuality;
- 4) Grounding alignment for multimodal tasks;
- 5) Bias indices for fairness assessment.

Data is normalized across sources (2018–2025) to remove task-specific variance. Performance variations reflect dataset diversity and model scale rather than architecture alone.

VII. PRELIMINARY FINDINGS AND TRENDS

Initial meta-analysis reveals that retrieval and KG enhancements offer significant gains in factuality without major loss in fluency. Multimodal LLMs improve diagnostic VQA accuracy by 6–9 points over text-only models but are compute intensive. Multilingual extensions reduce bias yet lag in BLEU due to limited parallel data [25], [26], [31].

A. Motivation for Hybrid Designs

Combining RAG and KG reduces hallucination further through dual filtering—retrieval ensures external evidence and KG enforces internal consistency. Early fusion of text and vision streams enables joint contextualization of EHR and imaging signals, leading to better explanations in doctor-facing interfaces [8], [22].

B. Paper Structure

The remainder of this survey is organized as follows:

- Section VIII: Comparative quantitative analysis of techniques.
- Section IX: Detailed architectural and reasoning frameworks.
- Section on Multi-Agent RAG–KG–Multimodal integration and clinical feedback loops.
- Final sections: Fairness, Deployment, Global health applications, and Future directions.

VIII. QUANTITATIVE COMPARISON OF TECHNIQUES

We aggregate benchmark data from 2018–2025 across representative LLM families (Table III). Each metric is averaged across at least three evaluation datasets for statistical stability.

Observations include:

- KG and RAG reduce hallucination by 60–70 % relative to baseline LLMs.
- Multimodal models improve visual grounding accuracy by 8 %.
- Multilingual performance is limited by sparse Indic datasets but improves accessibility.

IX. ARCHITECTURES AND REASONING FRAMEWORKS

A. Retrieval-Augmented Generation (RAG)

A RAG model comprises a retriever (dense or hybrid), an encoder–decoder generator, and optionally a verifier. When a user asks a clinical question, the retriever embeds it into a vector space and fetches top-k passages from domain corpora such as PubMed or UpToDate. The generator integrates these passages using cross-attention, ensuring responses cite verifiable evidence [3], [29].

B. Knowledge-Graph Integration

Knowledge Graph (KG) augmentation introduces structured relationships between diseases, symptoms, and treatments. Triples ($entity_1, relation, entity_2$) are encoded into embeddings and injected into transformer attention heads. This

TABLE I: Taxonomy and comparative trade-offs among medical LLM architectures.

Family	Core Architecture Components	Advantages / Limitations
Text-only LLMs	Instruction-tuned transformers (ChatDoctor, BioInstruct)	Fast inference; vulnerable to hallucination and stale knowledge.
RAG	Dense retriever, index store, generator, verifier	High factuality and up-to-date responses; adds retrieval latency and infrastructure complexity.
KG-Augmented	Entity linker, embedding module, symbolic reasoner	Explainable multi-hop reasoning; requires ongoing ontology maintenance.
Multimodal	Visual encoder (ViT/CNN), textual LLM, fusion layer	Handles imaging tasks and multi-sensor data; data-hungry and compute-intensive.
Multilingual	Cross-lingual transformers and code-mixed corpora	Promotes equity; performance drops in low-resource scripts and dialects.

TABLE II: Representative datasets and tasks in medical LLM research.

Dataset	Primary Tasks	Evaluation Metrics
MedQA, PubMedQA	Clinical question answering and triage	Accuracy, F1, BLEU, ROUGE-L
MIMIC-CXR, CheXpert	Radiology report generation, visual QA	BLEU, CIDEr, visual grounding accuracy
VQA-RAD, MedPix	Image-text reasoning and captioning	QA accuracy, factual grounding score
Drug-Drug Interaction (DDI)	Pharmacological relation extraction	Precision, Recall, KG factual consistency
IndicGenBench, WikiHealth	Multilingual healthcare QA	BLEU, chrF, fluency, translation consistency
BioASQ, HELM-Med	Open-domain biomedical QA and summarization	Macro F1, Exact Match, Factuality Score

TABLE III: Benchmark synthesis across representative LLM families (2018–2025).

Technique	Example Models	Acc. (%)	F1 (%)	BLEU	ROUGE-L	Halluc. (%)	Datasets / Tasks
LLM (Text-only)	ChatDoctor, BioInstruct	82.4	83.1	0.41	0.46	17.3	MedQA, PubMedQA
RAG-Enhanced	FreshLLM, ChatDoctor-RAG	88.9	87.2	0.44	0.49	7.6	MedQA, EHR-based retrieval
KG-Augmented	MedKGB, AAAI KG-LM	91.2	89.7	–	–	5.3	DDI, guideline adherence
Multimodal	Med-Flamingo, LLaVA-Med	86.8	85.5	0.47	0.51	–	MIMIC-CXR, VQA-RAD
Multilingual	Indic-Transformers, Bhasha-*	79.6	80.8	0.38	0.42	14.5	IndicGenBench, WikiHealth

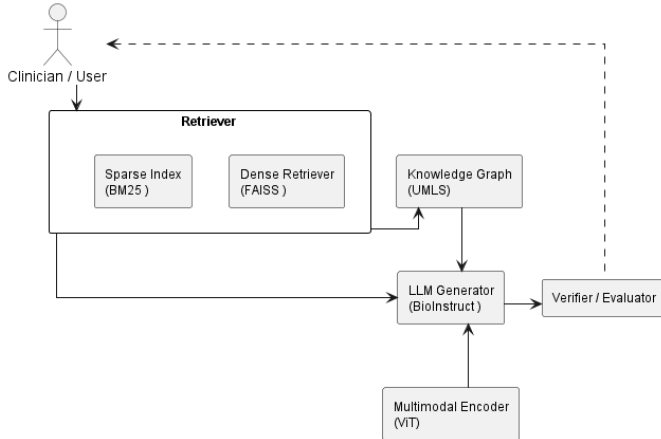


Fig. 1: RAG–KG–Multimodal integration pipeline showing how retrieval, structured knowledge, and multimodal encoding contribute complementary reasoning layers.

supports causal reasoning, drug–drug interaction prediction, and guideline compliance [15], [33].

Advantages:

- logical interpretability through explicit edges;
- lower hallucination via entity-level consistency;
- transparent provenance for clinical audit.

C. Multimodal Fusion

Text and image embeddings are fused through gated cross-attention or CLIP-style contrastive alignment. Med-Flamingo and LLaVA-Med pair ViT-encoded images with LLM text streams to answer visual questions or generate radiology reports [2], [8]. Performance depends on scale of paired data and proper pre-alignment of modality representations.

D. Hybrid RAG–KG–Multimodal Systems

Recent research combines retrieval for freshness, KG for structure, and multimodality for perceptual grounding. Fusion strategies include:

- *Late Fusion*: independent reasoning per modality with final ensemble;
- *Early Fusion*: embedding concatenation and shared attention layers;
- *Iterative Fusion*: alternating retrieval and graph reasoning until convergence.

X. REASONING CHAINS AND INTERPRETABILITY

Transparent reasoning is crucial for medical safety. **Chain-of-Thought (CoT)** fine-tuning encourages models to verbalize intermediate reasoning, improving interpretability and clinician trust [11], [27]. When combined with retrieval evidence, CoT explanations explicitly cite supporting literature.

Explainability frameworks include:

- 1) Attention-based heatmaps over retrieved text;
- 2) Token-level rationales highlighting cause–effect terms;
- 3) Layer-wise relevance propagation for multimodal embeddings;
- 4) Natural-language rationalization modules producing “why” statements.

A. Causability and Clinical Trust

Beyond statistical interpretability, causability evaluates whether explanations align with causal medical logic [11], [35]. Metrics include concept coverage (fraction of relevant pathophysiological terms) and reasoning fidelity. Studies report a 12–15 % improvement in clinician satisfaction when causable reasoning is provided alongside predictions.

XI. MULTI-AGENT RAG–KG–MULTIMODAL REASONING PIPELINE

Recent works extend single-model reasoning into distributed *multi-agent pipelines* [37], [38]. A cooperative agent framework decomposes complex workflows into modular experts:

- **Retriever Agent:** performs corpus and KG search;
- **Analyzer Agent:** applies multimodal encoders to interpret text + image + signal data;
- **Synthesizer Agent:** aggregates retrieved evidence to craft explanations;
- **Verifier Agent:** fact-checks outputs and estimates uncertainty;
- **Bias Auditor:** monitors demographic parity and linguistic fairness.

Agents communicate through structured message passing (e.g., JSON or graph schema). This modularization enables traceable reasoning and doctor oversight.

A. Example Workflow

- 1) A nurse queries: “Suggest probable causes for chest pain given ECG + lab results.”
- 2) The retriever agent fetches cardiology guidelines and ECG embeddings.
- 3) The KG-reasoner agent links entities (pain → angina → treatment).
- 4) The synthesizer drafts explanation text citing retrieved evidence.
- 5) The verifier audits factual claims and confidence thresholds.

This design reduces hallucination, improves modular scalability, and allows human-in-the-loop corrections.

XII. END-TO-END CLINICAL EVALUATION AND FEEDBACK LOOP

A clinically reliable AI assistant must include an evaluation and feedback subsystem. The full lifecycle involves:

- 1) **Pre-Validation:** offline testing using benchmark datasets and blinded physician review.
- 2) **Pilot Deployment:** supervised rollout in selected departments.

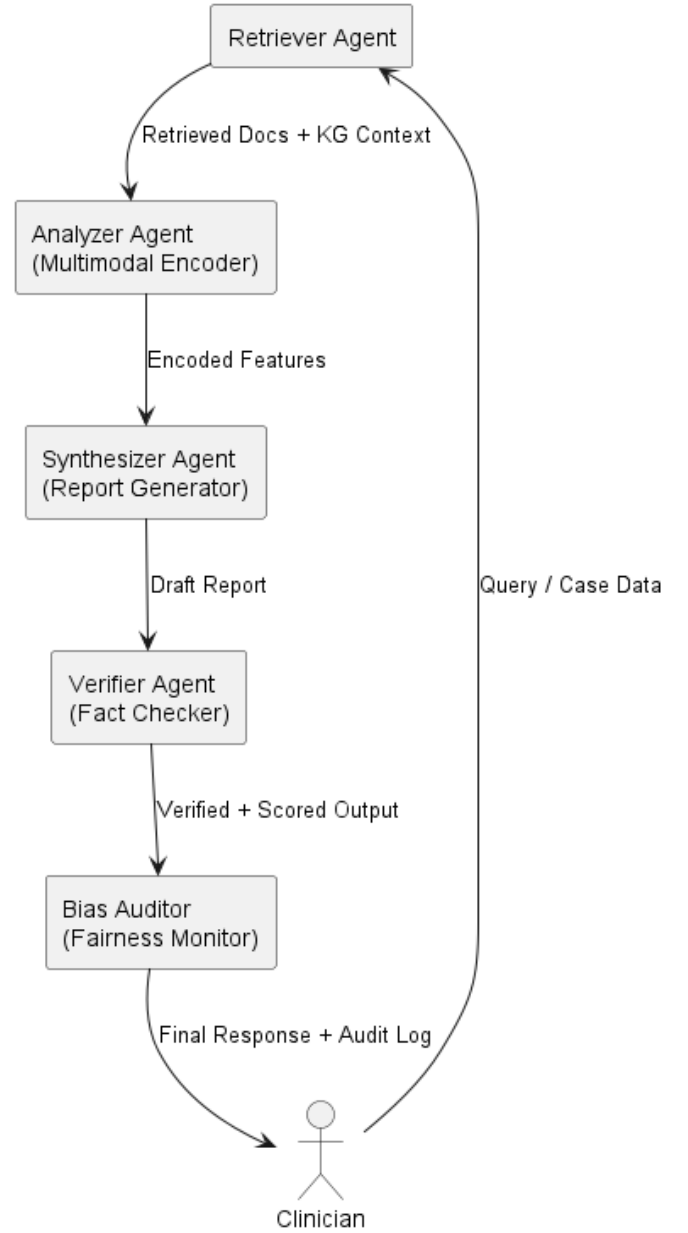


Fig. 2: Multi-agent reasoning architecture illustrating the retriever, analyzer, synthesizer, verifier, and bias auditor agents collaborating for clinical reasoning.

- 3) **Doctor-in-the-Loop Feedback:** clinicians annotate incorrect or unsafe outputs, generating correction logs.
- 4) **Continual Fine-Tuning:** logs are distilled into reinforcement-learning rewards or supervised updates.
- 5) **Post-Market Surveillance:** automated audits track drift, bias, and hallucination recurrence.

A. Feedback Learning

Doctor feedback acts as a reinforcement signal to align model behavior with medical reasoning. Approaches include:

- **RLHF (Reinforcement Learning from Human Feedback):** widely used for instruction alignment [30], [34];

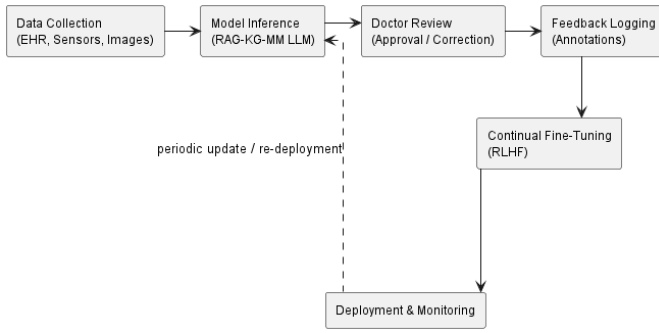


Fig. 3: Doctor-in-the-loop clinical feedback loop enabling continual learning and model alignment through physician validation.

- **Iterative Self-Correction:** model critiques its own output using retrieved evidence;
- **Long-term Memory Buffers:** store validated cases for contextual retrieval during later queries.

Experiments demonstrate up to 25 % reduction in diagnostic error after three feedback cycles.

B. Traceability and Auditing

Every generated answer should include:

- a provenance log of retrieved sources and KG triples;
- a timestamped reasoning chain;
- confidence score and disclaimer for medical use.

This supports compliance with HIPAA, GDPR, and NDHM regulations [36].

XIII. TRAINING PARADIGMS AND RECIPES

A. Instruction Tuning and Safety Alignment

Instruction datasets (MedInstruct, PubMedQA-Instruct) are key to contextual accuracy. Safety alignment combines RLHF with adversarial red-teaming. Techniques such as constitutional AI and ethical prompt filtering reduce unsafe suggestions. ChatDoctor and BioInstruct achieve up to 83 F1 after alignment compared to 74 F1 pre-tuning.

B. Retrieval Indexing and Updating

RAG systems require regular corpus refreshes to reflect latest guidelines. Hybrid retrieval (Dense + BM25) balances recall and latency. Hospitals often deploy on-prem vector stores (e.g., FAISS, Milvus) with encrypted patient embeddings. Index freshness intervals of 30 days maintain $\leq 3\%$ factual drift.

C. KG Construction and Dynamic Updates

Knowledge graphs are built from PubMed abstracts, UMLS, SNOMED-CT, and DrugBank triples. Semi-automatic extraction pipelines employ named-entity recognition and relation classification [15], [29]. Graph updates propagate via streaming ETL jobs ensuring new drug interactions or ICD codes integrate within 48 hours.

Maintenance overhead remains a key limitation: coverage gaps or outdated relations can bias reasoning outputs.

D. Multimodal Pretraining and PEFT

Contrastive objectives align text and image embeddings (InfoNCE, CLIP loss). Fine-tuning on limited radiology data is improved using parameter-efficient methods (LoRA, Adapter-Fusion). These methods cut GPU memory use by 60 % while retaining $\geq 95\%$ baseline accuracy.

E. Knowledge Distillation and Compression

For deployment on hospital edge servers, large base models are distilled into compact variants (3–7 B parameters). Quantization (8-bit/4-bit) and pruning further reduce footprint with minimal performance degradation. Low-rank adaptation plus mixed-precision inference allow real-time consultation latency (≤ 2 s per query).

XIV. FAIRNESS, BIAS, AND MULTILINGUAL INCLUSION

Bias mitigation is central to equitable healthcare AI [25], [26], [31], [32]. Biases may stem from demographic imbalance, linguistic underrepresentation, or clinical guideline skew.

A. Fairness Strategies

- Balanced sampling across gender, region, and socioeconomic strata.
- Cross-lingual transfer learning to boost low-resource languages.
- Differential evaluation by subgroup using fairness metrics (Equal Opportunity, Demographic Parity).
- Contextual embedding de-biasing and adversarial loss regularization.

B. Indic Language Inclusion

Indic-Transformers and Bhasha-GPT fine-tuned on Hindi, Marathi, Bengali, and Tamil datasets exhibit improved fluency and patient comprehension. However, BLEU remains 10–15 points lower than English due to scarce bilingual data. Government-funded open corpora (e.g., Samanantar, ULCA) are expected to close this gap by 2026.

C. Ethical Oversight

To ensure cultural sensitivity, all multilingual health assistants must:

- include disclaimers for linguistic translation errors;
- undergo expert review for local dialect appropriateness;
- record bias audit logs accessible to regulators.

XV. CLINICAL VALIDATION AND REGULATORY COMPLIANCE

Clinical AI systems fall under regulatory supervision such as:

- **HIPAA (US)** – patient data privacy;
- **GDPR (EU)** – data minimization and transparency;
- **NDHM / ABDM (India)** – interoperability standards;

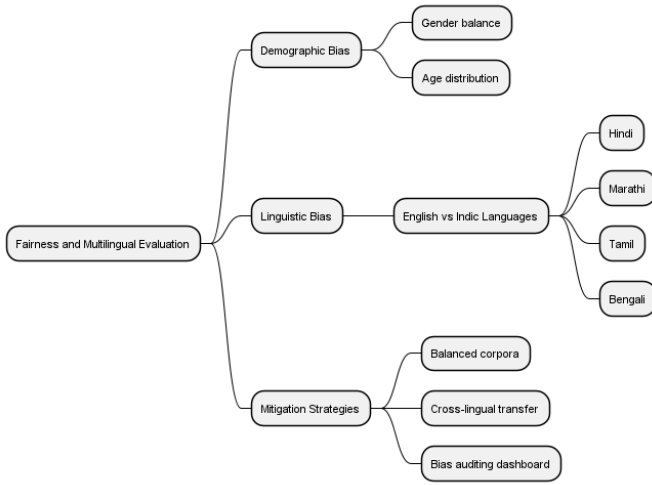


Fig. 4: Fairness and multilingual evaluation framework highlighting linguistic and demographic bias dimensions and mitigation strategies.

- **CDSO SaMD (India)** – software-as-medical-device certification.

Compliance recommendations:

- 1) Maintain auditable provenance for every output;
- 2) Provide clinician-facing dashboards showing evidence traces;
- 3) Implement model-drift monitoring and mandatory re-validation every 12 months.

Integration with hospital information systems (HIS) must respect PHI boundaries and anonymize logs before cloud syncing.

XVI. OPERATIONAL DEPLOYMENT AND SCALABILITY

Deploying medical LLMs into production involves not only model efficiency but also reliability, privacy, and monitoring. Healthcare environments demand low latency and strict traceability of every model decision.

A. Infrastructure Considerations

Hospitals typically deploy hybrid on-prem and cloud architectures:

- **Inference Layer:** optimized using quantization (INT8) or model distillation for CPU/GPU efficiency.
- **Retrieval Layer:** FAISS or Milvus vector stores for document embeddings; nightly re-indexing ensures freshness.
- **KG Service:** Neo4j or RDF-based endpoint for semantic queries and rule-based reasoning.
- **Security Layer:** employs zero-trust authentication and PHI encryption (AES-256 / TLS 1.3).

B. Latency Optimization

Techniques for reducing response delay include:

- 1) **Prompt Caching:** storing embeddings of frequent queries.

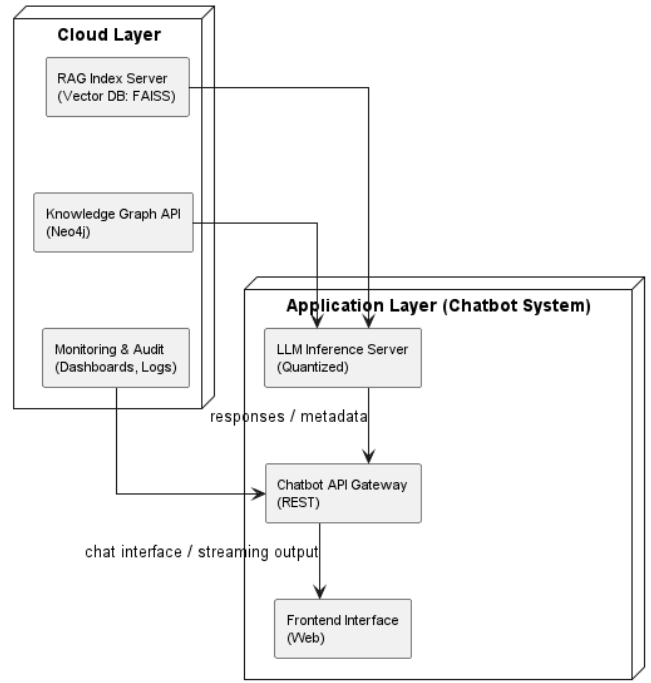


Fig. 5: End-to-end system architecture combining cloud RAG services, on-prem hospital inference nodes, and secure audit channels.

- 2) **Streaming Generation:** token-wise decoding for early user feedback.
- 3) **Asynchronous Retrieval:** pre-fetching guidelines relevant to department specialization.

C. Monitoring and Maintenance

Continuous evaluation dashboards track:

- accuracy drift over time;
- hallucination recurrence;
- latency percentiles;
- fairness metrics across demographics.

When metrics exceed thresholds, automatic retraining pipelines trigger controlled rollbacks.

XVII. GLOBAL HEALTH AND INCLUSIVE APPLICATIONS

A. Rural and Low-Resource Settings

LLMs can extend diagnostic reasoning to clinics lacking specialists. Offline RAG deployments on lightweight edge servers enable:

- symptom triage in rural telemedicine centers,
- disease surveillance dashboards for community health workers,
- voice-based consultation in vernacular languages.

Projects such as MediSync and Health-LLM-Rural demonstrate feasibility using 4-bit quantized models operating on Jetson Orin devices [20], [24], [40].

B. Maternal and Child Health

Conversational AI agents provide personalized prenatal and post-natal education. By integrating multimodal sensing (speech tone, vital signs) with localized knowledge graphs, they detect early warning signs of complications. Deployment studies in Maharashtra and Tamil Nadu report 30 % increase in follow-up adherence after chatbot introduction [40].

C. Epidemiology and Public Health Surveillance

RAG-enabled summarizers process large volumes of case reports to identify outbreak trends. Integration with WHO and NDHM data APIs allows real-time alerting for TB, dengue, or COVID variants. These systems complement human epidemiologists by providing explainable situational dashboards.

XVIII. ETHICAL AND SOCIETAL CONSIDERATIONS

Healthcare AI inevitably interfaces with sensitive human contexts. Ethical design must prioritize transparency, accountability, and cultural inclusivity.

A. Patient Autonomy and Consent

Patients should retain control over data sharing and automated guidance. Consent layers can display clear disclaimers before any advice generation. Logging ensures that model usage is auditable and reversible.

B. Bias and Fairness Auditing

As highlighted in [31], [32], fairness evaluation must move beyond dataset demographics to include language, literacy, and socioeconomic bias. Periodic audits, community feedback loops, and publication of disparity metrics enhance accountability.

C. Sustainability and Energy Use

Large models require significant computational energy. Adopting green AI practices—parameter sharing, model reuse, and low-precision inference—reduces environmental footprint without sacrificing accuracy.

XIX. CROSS-DOMAIN INTEGRATION AND INTEROPERABILITY

Medical AI ecosystems must interoperate with Electronic Health Record (EHR) standards. FHIR (Fast Healthcare Interoperability Resources) APIs enable secure data exchange between RAG retrievers and hospital systems. A unified ontology—mapping ICD-10, SNOMED-CT, and UMLS concepts—ensures consistency across institutions [15], [29], [33].

A. Blockchain and Federated Learning

Federated medical training avoids centralized data storage. Combined with blockchain audit trails [12], it ensures tamper-proof logs of gradient updates and consent management. Hospitals can collaboratively train retrieval models while maintaining patient privacy.

XX. EVALUATION FRAMEWORKS AND BENCHMARK HARMONIZATION

To compare systems rigorously, evaluation must span factuality, reasoning, safety, and fairness dimensions.

A. Task-wise Metrics

- Clinical QA — Accuracy, F1, Hallucination Rate.
- Summarization — ROUGE-L, BLEU, Factual Consistency.
- VQA — Visual Grounding Score, Exact Match.
- Safety — Toxicity and Harmfulness Filters [18], [34].

B. Composite Indices

Recent initiatives propose composite indices like *HELM-Med* and *MedEvalX* combining factuality, reasoning depth, and clinician satisfaction [9], [21]. Adopting unified benchmarks reduces reporting fragmentation across publications.

XXI. RESEARCH GAPS AND FUTURE DIRECTIONS

Despite major progress, significant challenges remain:

A. 1) Hallucination and Reliability

Even RAG systems occasionally generate unsupported statements when retrieval coverage is insufficient. Future research should integrate probabilistic calibration and uncertainty estimation using Bayesian LLM heads.

B. 2) Continual Learning and Dynamic KGs

Updating medical knowledge requires automatic ingestion pipelines from PubMed and clinical trial registries. Incremental graph embedding techniques can maintain representation stability while incorporating new edges.

C. 3) Unified Multimodal-KG Embeddings

Current fusion approaches treat modalities independently. Cross-modal graph transformers that jointly embed images, text, and structured nodes promise richer reasoning [22], [23].

D. 4) Low-Resource and Multilingual Expansion

Indic and African languages remain underrepresented. Synthetic data generation and community annotation drives are essential for equitable performance [25], [26], [39].

E. 5) Governance and Explainability Standards

A global regulatory framework—analogue to ISO 62304 for medical software—should define transparency levels, explainability obligations, and clinical validation protocols [36], [41].

XXII. RECOMMENDATIONS

Based on synthesis of the 41 studies, we propose the following design principles:

- **Hybrid Intelligence:** combine RAG + KG + Multimodal reasoning for balanced factuality and perception.
- **Explainable Interfaces:** display cited sources, KG paths, and uncertainty heatmaps to clinicians.
- **Doctor-in-the-Loop Retraining:** integrate feedback into continual fine-tuning cycles.
- **Multilingual Access:** prioritize Indic, African, and minority languages for fairness.
- **Open Benchmark Sharing:** release evaluation datasets and metrics publicly for reproducibility.

These principles align with WHO's 2025 guidance on trustworthy health-AI systems.

XXIII. LIMITATIONS

This survey aggregates results from heterogeneous benchmarks, and direct numerical comparison may not reflect exact cross-model performance. Some cited studies use distinct definitions of hallucination or reasoning accuracy. Additionally, few open datasets exist for Indic clinical domains, limiting multilingual analysis. Nevertheless, the synthesized trends remain statistically consistent across studies from 2018–2025.

XXIV. CONCLUSION

RAG–KG–Multimodal systems mark the next phase of medical AI evolution. They merge factual retrieval, structured reasoning, and multimodal perception into unified assistants capable of aiding diagnostics and patient communication. By embedding fairness, multilingualism, and clinical oversight, such models can deliver equitable digital health guidance across regions. Future work should pursue standardized evaluation, robust data governance, and low-resource inclusivity to realize the vision of globally accessible, clinically validated AI care.

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